

LETS - a Wearable Heart and Lung Monitoring Device for the Diagnosis of Cardiac and Respiratory Diseases

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ABSTRACT

The paper reports progress on a wearable device for real-time monitoring of a patient's lungs and heart. Using a set of electrodes, we can measure various electrical potentials on the patient's body, which can be used to extract valuable medical information. In addition, electrical tomography and body surface potential mapping have been used for signal analysis and image reconstruction.

CCS CONCEPTS

• **Hardware** → Printed circuit boards; • **Human-centered computing** → Visualization design and evaluation methods; • **Applied computing** → Life and medical sciences.

KEYWORDS

electrical impedance tomography; medical imaging; wearable device

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1 INTRODUCTION

Cardiovascular and pulmonary diseases are a civilization problem that humanity will face in this century. Ischemic heart disease, strokes, and COPD are the infamous podium of diseases most often responsible for preventable premature deaths in developed countries [1].

Technological advances are making it possible to develop new wearable devices capable of collecting medical-grade data, from simple wristbands that count steps and measure heart rate to specialized devices designed for diagnostics. Motivated by these needs, we decided to develop LETS (Lung Electrical Tomography System) for comprehensive diagnostics of both the cardiovascular and respiratory systems. The device is intended to be inexpensive (up to several thousand Euro) and give more diagnostic information than existing equipment.

In this short article, we describe the current status of the development of a wearable device for measuring heart and lung function in the form of a vest. Our device is intended for professional use under a doctor's order. It is planned to be able to collect data over a 24-hour period, which would complement standard ECG Holter tests.

2 HARDWARE

On the hardware side, our device utilizes several methods for collecting data from the patient. These are body surface potential mapping (BSPM), impedance cardiography and electrical impedance

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Figure 1: The medical vest for conducting comprehensive lung and heart measurement.

tomography. The usage of these techniques together allows for multichannel monitoring of heart and respiratory action. The acquired data can be helpful in the diagnosis of diseases such as COPD, ARDS, hypertension or CAD.

The first two methods are intended for cardiac monitoring. BSPM extends the capabilities of classic ECG by conducting measurements on 102 electrodes. It offers new possibilities for diagnosing heart diseases, especially when data is processed by models based on

artificial intelligence [2]. Panel (a) on Figure 1 shows a picture of the inside of the vest with the textile electrodes visible. The EIT measurement is conducted with two electrode rings below the armpit level, each containing 16 electrodes. On the other hand EIT, allows for thorax imaging. This can be extremely useful for tracking the lung volume vs. time characteristic, which can serve as an input for sickness classification models.

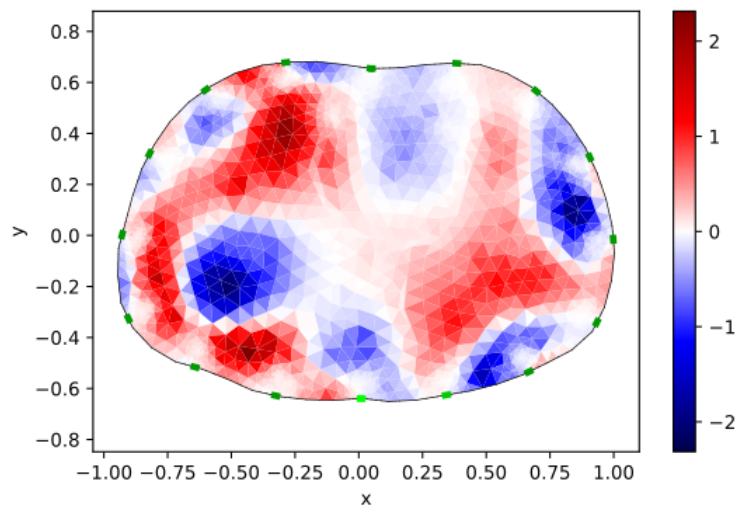


Figure 2: A sample reconstruction (relative impedance) of the thorax interior.

3 CLINICAL TRIALS

To certify and verify the device's operation, it was necessary to conduct clinical trials. To this point, the first phase of the study has been completed. It involves testing 60 randomly selected patients. For the second planned phase, the group will be properly selected regarding the appropriate proportion of healthy and sick people. The test procedure included, i. a., measurements during rest and light exercises. The clinical trial provides large amount of data that was processed with the expert system developed during product development. The main goal of the expert system is to detect any disease that are connected with cardio-pulmonary system.

4 EARLY RESULTS AND RECONSTRUCTIONS

This section will discuss an example of an EIT reconstruction result successfully obtained during a clinical trial. Data from the lower ring of the EIT, located below the patient's nipple line, was selected for its performance.

The reconstructed image can be seen in Figure 2. The colours in the image represent the impedance relative to a fixed background. For example, the lungs are visible on the left and right sides of the image (red). On the other hand, the heart is visible in the upper part of the image (blue).

The inverse problem (reconstruction of the image from the voltage measurement) was solved using the Gauss-Newton method with a regularization constant of $\lambda = 1.5 \times 10^{-6}$ on a triangular grid with 2914 elements. The current excitation was between the first and eighth electrodes - the centre of the back and the centre of the chest, respectively.

5 CONCLUSION

In this paper, we report our recent progress on the wearable lung and heart monitoring device for comprehensive diagnostics of respiratory and circulatory diseases. We present working EIT imaging, which successfully can detect a patient's lungs and heart. During clinical trials, the device proved to be useful for monitoring the performance of heart during the workout and post-workout periods.

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